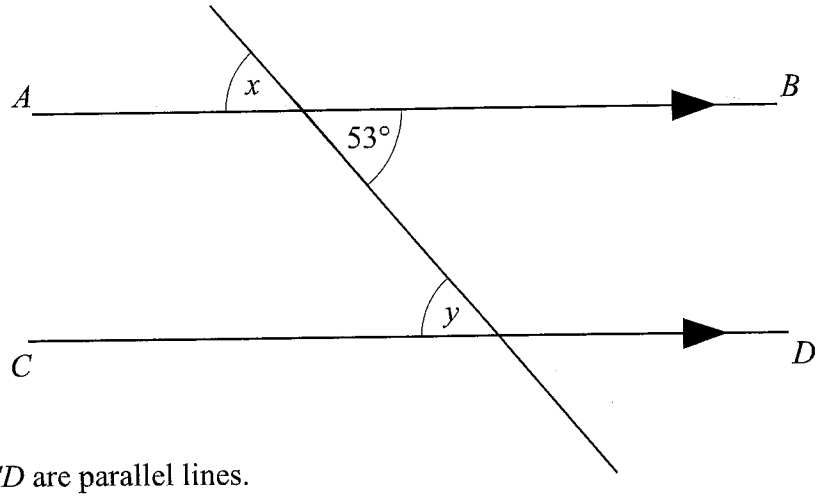
1



AB and *CD* are parallel lines.

(a) Write down the size of angle *x*.

53 °

(1)

(b) Give a reason for your answer.

(vertically) opposite angles are equal

(1)

(c) Write down the size of angle *y*.

53 °

(1)

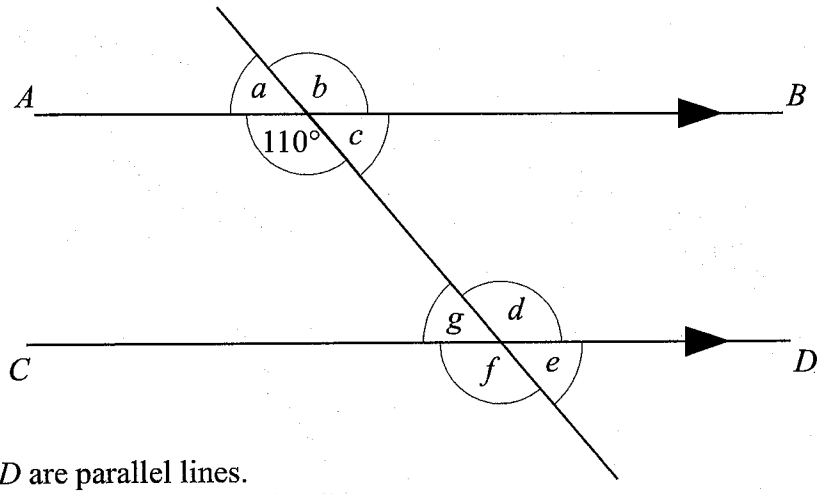
(d) Give a reason for your answer.

alternate angles are equal

(1)

(Total for question 1 is 4 marks)

2



AB and *CD* are parallel lines.
An angle of 110° is shown on the diagram.

(a) Write down the letter of one other angle of size 110°

.....f.....
(1)

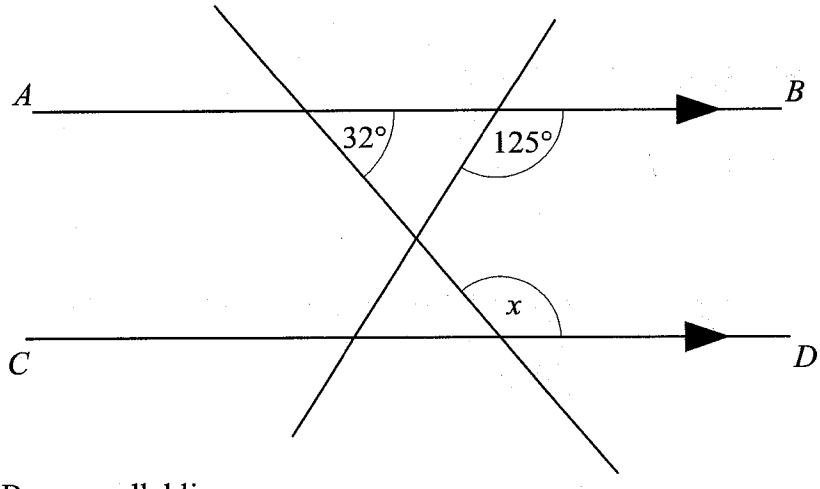
(b) Give a reason for your answer.

...Corresponding angles are equal...

[OR// ...b: opposite angles are equal d: alternate angles equal]
(2)

(Total for question 2 is 3 marks)

3



AB and *CD* are parallel lines.

(a) Find the size of angle *x*

180 − 32 = 148

.....148°.....
(1)

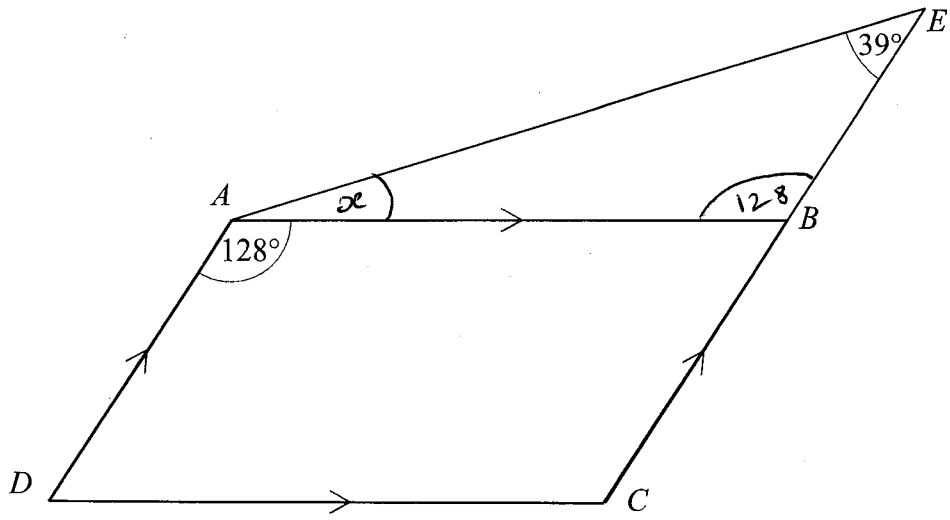
(b) Give a reason for your answer.

...Co-interior angles add to 180°...

[OR// angles on a straight line add to 180° AND
Corresponding / alternate angles are equal]
(2)

(Total for question 3 is 3 marks)

4



ABCD is a parallelogram.
CBE is a straight line.
Angle *BAD* = 128°
Angle *AEB* = 39°

Find the size of angle *BAE*.
Give a reason for each stage of your working.

Angle ABE = 128° Alternate angles are equal

128 + 39 = 167°

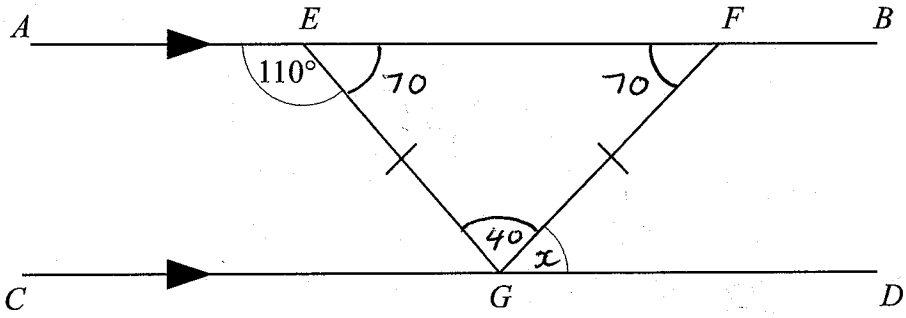
180 − 167 = 13°

BAE = 13° Angles in a triangle add to 180°

13 .....°

(Total for question 4 is 3 marks)

5



AB and *CD* are parallel lines.
EFG is an isosceles triangle

Angle *AEG* = 110°

Find the size of angle *FGD*.
Give a reason for each stage of your working.

Angle FEG = 180 − 110
= 70°

Angle EFG = 70° Angles at the base of an isosceles triangle are equal

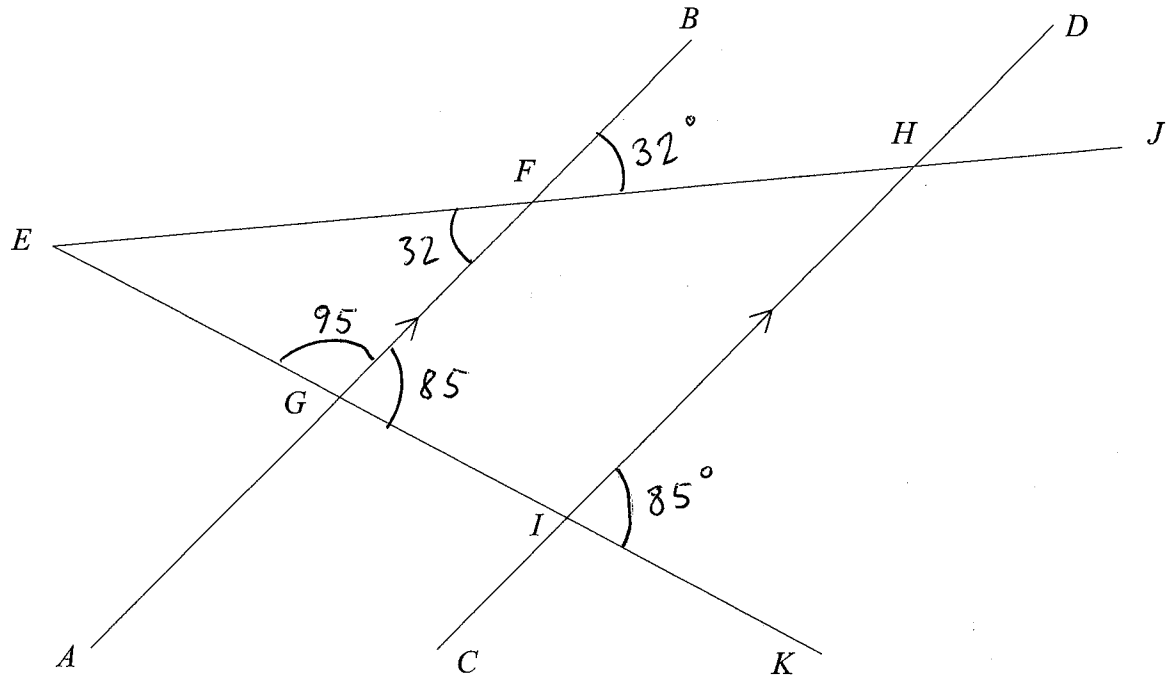
~~Angle EGF = 180 − 70 − 70 Angles in a triangle add to 180°~~
~~= 40~~

FGD = 70° Alternate angles are equal

70 °

(Total for question 5 is 3 marks)

6



AB and *CD* are parallel.
Angle *HIK* = 85°
Angle *BFH* = 32°

Find the size of angle *FEG*.
You must show how you got your answer.

EFG = 32° opposite angles are equal

FGI = 85° corresponding angles are equal

EGF = 95° Angles on a straight line add to 180°.
(180 − 85 = 95)

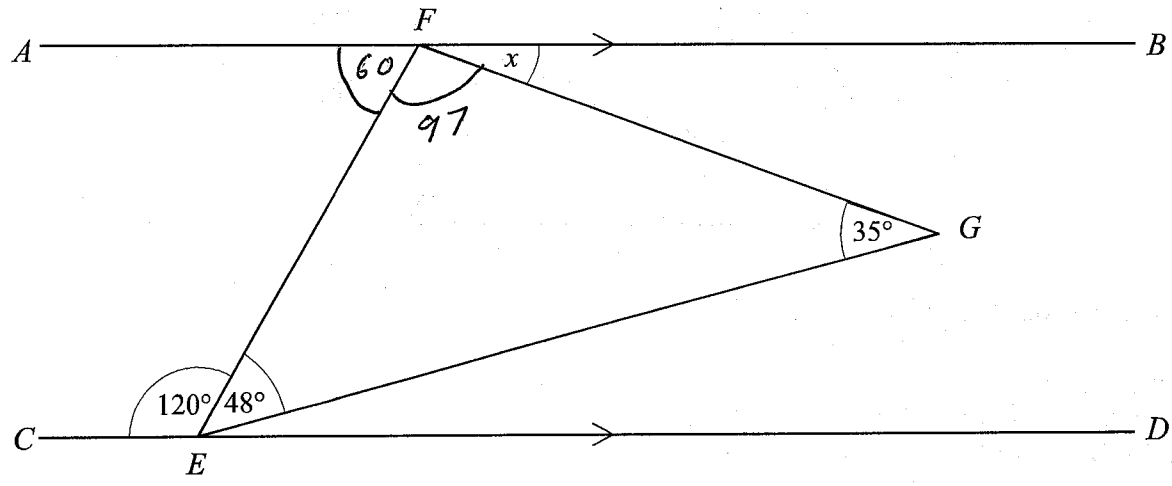
180 − 95 − 32 = 53°

FEG = 53° Angles in a triangle add to 180°

.....53.....°

(Total for question 6 is 3 marks)

7



AB and *CD* are parallel.

Find the size of angle *x*.
Give a reason for each stage of your working.

180 − 120 = 60

AFE = 60° Co Interior angles add to 180°

EFG = 180 − 48 − 35 = 97°

Angles in a triangle add to 180°

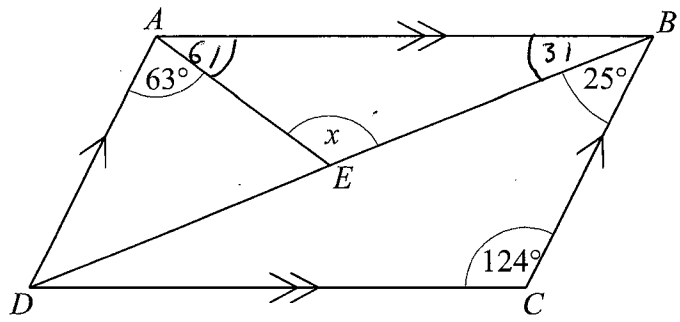
180 − 97 − 60 = 23°

x = 23° Angles on a straight line add to 180°

.....23.....°

(Total for question 7 is 4 marks)

8



ABCD is a parallelogram.

Angle *DAE* = 63°
Angle *BCD* = 124°
Angle *CBD* = 25°

Calculate the size of angle *x*.
Give reasons for each stage of your answer.

opposite angles in a parallelogram are equal

∴ BAD = 124°

BAE = 124 − 63 = 61°

ABC = 180 − 124 = 56° Co Interior angles add to 180°

ABE = 56 − 25 = 31°

180 − 61 − 31 = 88° Angles in a triangle add to 180°

.....88.....°

(Total for question 8 is 3 marks)